INCH-POUND

MIL-W-22759/4C
10 April 1992
SUPERSEDING
MIL-W-22759/4B
10 September 1976

MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, TFE-GLASS-FEP, MEDIUM WEIGHT, SILVER-COATED COPPER CONDUCTOR, 600-VOLT

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-22759.

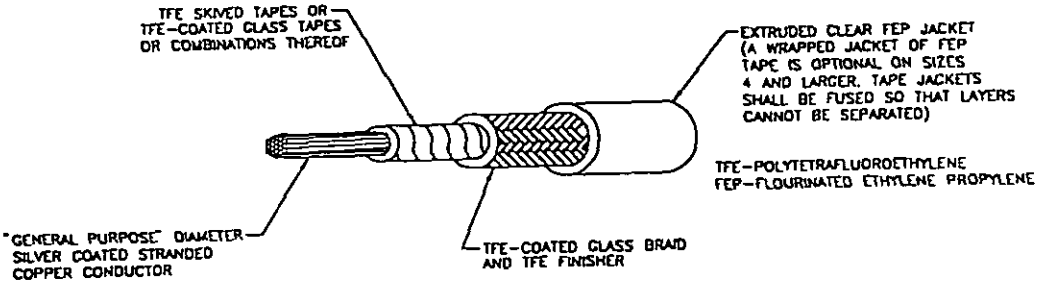


TABLE I. Construction details.

Part No. 1/	Wire size	Stranding (Number of strands x AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire			
			(min)	(max)	Resistance at 20°C(68°F) (ohms/1000 ft) (max)	Thickness of FEP jacket (inches)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/4-22-*	22	19 x 34	.029	.032	15.1	.009 ±.002	.074 ±.003	5.90
M22759/4-20-*	20	19 x 32	.037	.040	9.19	.009 ±.002	.082 ±.003	7.90
M22759/4-18-*	18	19 x 30	.046	.050	5.79	.010 ±.002	.095 ±.003	11.0
M22759/4-16-*	16	19 x 29	.052	.057	4.52	.010 ±.002	.103 ±.004	13.6
M22759/4-14-*	14	19 x 27	.065	.072	2.88	.010 ±.002	.116 ±.004	18.5
M22759/4-12-*	12	37 x 28	.084	.089	1.90	.010 ±.002	.133 ±.004	26.6
M22759/4-10-*	10	37 x 26	.106	.112	1.19	.010 ±.002	.164 ±.006	41.5
M22759/4-8-*	8	133 x 29	.158	.169	.658	.010 ±.002	.235 ±.007	77.4
M22759/4-6-*	6	133 x 27	.198	.213	.418	.010 ±.002	.282 ±.010	115.
M22759/4-4-*	4	133 x 25	.250	.268	.264	.012 ±.003	.351 ±.015	184.
M22759/4-2-*	2	665 x 30	.320	.340	.170	.012 ±.003	.430 ±.015	281.
M22759/4-1-*	1	817 x 30	.360	.380	.139	.014 ±.004	.480 ±.015	358.
M22759/4-01-*	0	1045 x 30	.395	.425	.108	.014 ±.004	.525 ±.025	436.
M22759/4-02-*	00	1330 x 30	.440	.475	.085	.014 ±.004	.585 ±.025	554.

1/ PART NO: The asterisks in the part number column, tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/4-20-9; white with orange stripe - M22759/4-20-93.

(C) Denotes changes

AMSC N/A
FSC 6145

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TABLE II. Performance details.

Part No.	Bend testing			Test load (lbs) (±3%)	
	Mandrel diameter (inches) (±3%)				
	Life cycle (oven and bend tests) 1/	Cold bend test	Wrap test	Life cycle (oven and bend tests) 1/	Cold bend test
M22759/4-22-*	.375	.50	.250	.75	2.0
M22759/4-20-*	.375	.50	.250	.75	2.0
M22759/4-18-*	.50	.75	.375	1.0	2.0
M22759/4-16-*	.50	.75	.375	1.0	3.0
M22759/4-14-*	.50	1.00	.50	1.0	3.0
M22759/4-12-*	.75	2.00	.50	3.0	3.0
M22759/4-10-*	1.00	3.00	.75	3.0	5.0
M22759/4-8-*	1.50	3.00	1.00	3.0	5.0
M22759/4-6-*	2.00	4.00	1.25	6.0	10.0
M22759/4-4-*	3.00	6.00	1.50	6.0	10.0
M22759/4-2-*	4.00	6.00	2.00	6.0	15.0
M22759/4-1-*	6.50	10.00	2.50	6.0	15.0
M22759/4-01-*	10.00	10.00	3.00	10.0	20.0
M22759/4-02-*	10.00	18.00	4.00	10.0	25.0

1/ Also for bend tests after immersion.

REQUIREMENTS:

Temperature rating: 200°C (392°F) max conductor temperature.

Voltage rating: 600 volts (rms) at sea level.

Spark test of primary insulation: Not required.

Impulse dielectric test: 8.0 kilovolts (peak), 100 percent test.

Insulation resistance: 5,000 megohms for 1000 ft (min).

Wrap test: Mandrel test required - no cracking.
Dielectric test after mandrel wrap, 2500 volt (rms), 60Hz.

Blocking: 200 ±2°C (392 ±3.6°F).

Shrinkage: 0.125 inch max at 200 ±2°C (392 ±3.6°F).

Wicking: Size 22 through 12 - Procedure II; 2.0 percent (max) weight increase, 0.750 inch (max) dye travel.
Size 10 through 00 - No requirement.

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Low temperature (cold bend):
Bend temperature - -65 ±2°C (-85 ±3.6°F).
Dielectric test, 2500 volts (rms), 60Hz.

Thermal shock: Oven temperature, 200 ±2°C (392 ±3.6°F).
Max change in measurement - Sizes 22 through 12 - 0.060 inch.
Sizes 10 through 8 - 0.100 inch.
Sizes 6 through 00 - 0.125 inch.

Flammability: Post-flame dielectric test not required.

Life cycle: Oven temperature 250 ±2°C (482 ±3.6°F).
Dielectric test, 2500 volts (rms), 60Hz.

Dielectric test after immersion: 2500 volts (rms), 60Hz.

Humidity resistance: 5,000 megohms for 1000 ft, min insulation resistance after humidity exposure.

Surface resistance: Sizes 22 through 12 - 500 megohm-inches (min), initial and final readings.
Sizes 10 through 00 - No requirement.

© Smoke: 200°C (392°F).

Acid resistance: Required.
Dielectric test, 2500 volts (rms), 60Hz.

Color: In accordance with MIL-STD-104, Class 1; white preferred.

© Color striping or banding durability: 250 cycles (500 strokes) (min), 500 grams weight. Test not required if marking is under clear FEP jacket.

© Identification durability: 125 cycles (250 strokes) (min), 500 grams weight. Test not required if marking is under clear FEP jacket.

Wire length requirements: Schedule A.

Supersession data: The wire of this specification sheet, by part number, replaces and supersedes the wire of MS90294(AS) (canceled) in accordance with table III.

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TABLE III. Supersession by part number.

Part number MS90294(AS)	Part number MIL-W-22759/4
MS90294-22-*	M22759/4-22-*
MS90294-20-*	M22759/4-20-*
MS90294-18-*	M22759/4-18-*
MS90294-16-*	M22759/4-16-*
MS90294-14-*	M22759/4-14-*
MS90294-12-*	M22759/4-12-*
MS90294-10-*	M22759/4-10-*
MS90294-8-*	M22759/4-8-*
MS90294-6-*	M22759/4-6-*
MS90294-4-*	M22759/4-4-*
MS90294-2-*	M22759/4-2-*
MS90294-1-*	M22759/4-1-*
MS90294-01-*	M22759/4-01-*
MS90294-02-*	M22759/4-02-*

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - ME, MI
Navy - EC, OS
Air Force - 99
DLA - IS

User activities:
Army - AT, AV
Navy - MC

Preparing activity:
Navy - AS

Agent:
DLA-IS

(Project 6145-1152-17)

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